DSA Assignment

UNIT 1:

1. A company stores employee IDs in ascending order. Write a C program that accepts n employee IDs, searches for a required ID using Binary Search, displays its position when found, reports when it is absent, and counts the number of comparisons. Test the program for both successful and unsuccessful searches.

Source code :-

```
#include <stdio.h>

int main(){
   int a[100], n, key;
   int low, high, mid, comparisons = 0;
   int found = 0;
   printf("Enter number of employees: ");
   scanf("%d", &n);
   printf("Enter employee IDs in ascending order:\n");
   for (int i = 0; i < n; i++)
      scanf("%d", &a[i]);
   printf("Enter employee ID to search: ");
   scanf("%d", &key);
   low = 0;
   high = n - 1;
   while (low <= high)
   {
      mid = (low + high) / 2;
      comparisons++;
      if (a[mid] == key)
      {
         printf("Employee ID found at position %d\n", mid + 1);
         found=1;
  break;
      }
```

```
        else if (key < a[mid])
            high = mid - 1;
        else
            low = mid + 1;
    }
    if (found == 0)
        printf("Employee ID not found\n");
    printf("Number of comparisons: %d\n", comparisons);
    return 0;
}
```

Output :-

```
Enter number of employees: 5
Enter employee IDs in ascending order:
1 2 3 4 5
Enter employee ID to search: 3
Employee ID found at position 3
Number of comparisons: 1
```

```
Enter number of employees: 4
Enter employee IDs in ascending order:
101 103 105 107
Enter employee ID to search: 9
Employee ID not found
Number of comparisons: 2
```